

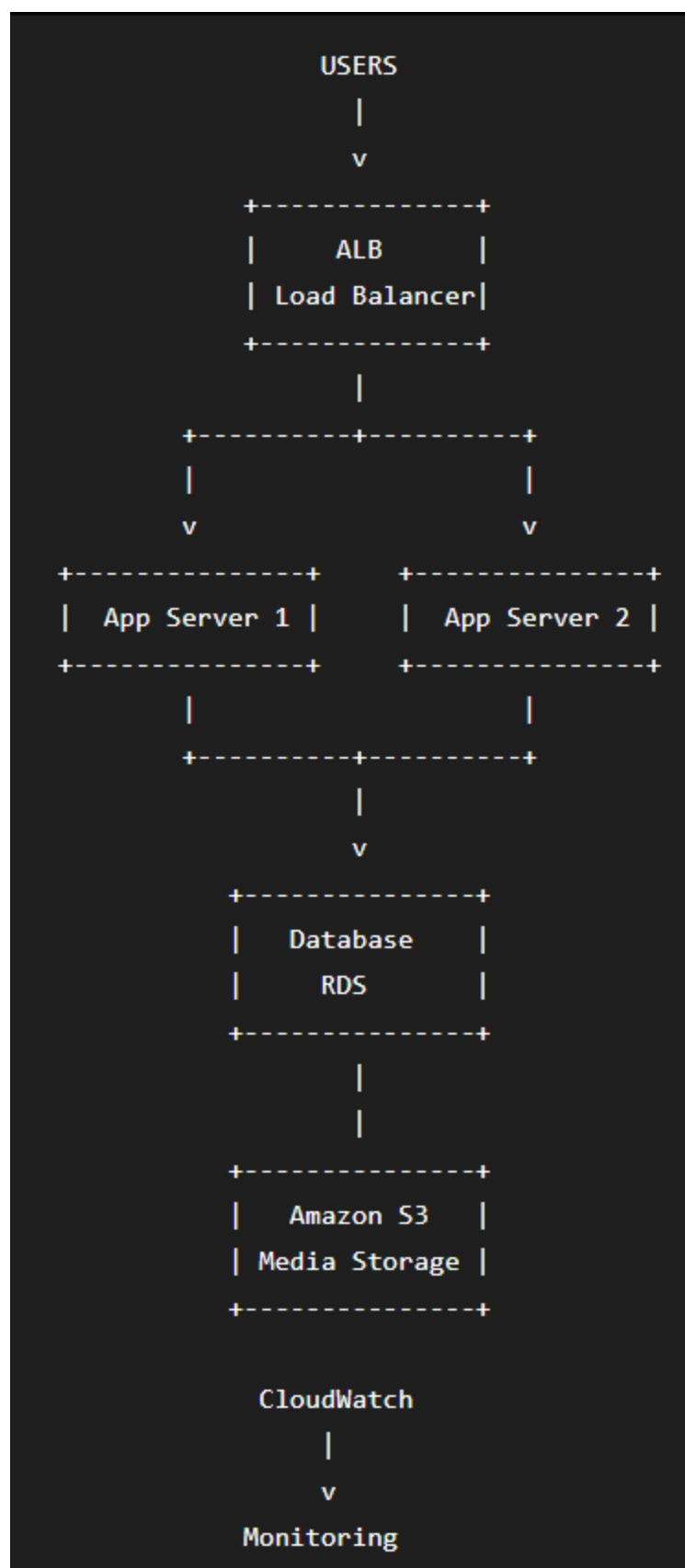
Task 1 — Selected App and Feature

Proposed Feature: Live Concert Streaming

I would add a Live Concert Streaming feature to Spotify that allows artists to stream live concerts to large numbers of users. The feature would require cloud hosting because the number of listeners could increase rapidly when a popular artist starts a live event.

The system would require scalable application servers, load balancing, databases, object storage and monitoring.

Task 2 — Cloud Architecture Diagram



Component Functions

Component	Purpose
Users	Access the Spotify application
ALB	Distributes requests
App Server 1	Processes application requests
App Server 2	Provides additional capacity
Database	Stores application/user information
S3	Stores media and other objects
CloudWatch	Monitors infrastructure

Task 3 — Security Measures

1. HTTPS/TLS

HTTPS encrypts communication between users and the application, reducing the risk of attackers intercepting sensitive information.

2. Security Groups and Firewalls

Security Groups restrict which network connections can reach application and database resources. Database servers should not be directly exposed to the public Internet.

3. Authentication and IAM

Strong authentication, MFA and least-privilege IAM policies reduce unauthorized access to applications and cloud resources.

Task 4 — 200–300 Word Architecture Explanation

Scalable Spotify Live Concert Architecture

The proposed architecture is designed to support a new live concert streaming feature for Spotify. Users first connect to the application through an Application Load Balancer (ALB). The load balancer distributes incoming requests between multiple application servers, preventing a single server from becoming a bottleneck.

During a major concert, the number of users may suddenly increase from thousands to millions. An Auto Scaling Group can monitor application or infrastructure metrics and automatically launch additional application servers when demand increases. When the event finishes and traffic decreases, unnecessary servers can be removed to reduce resource consumption and cost.

The application servers communicate with a managed database for storing user and application information. Database access is restricted so that it is not directly available from the public Internet. Amazon S3 can be used for storing media files, images and other large objects.

Security Groups and network controls restrict communication between different components. HTTPS protects communication between users and the application, while IAM policies enforce least-privilege access to AWS resources. CloudWatch can monitor CPU utilization, application metrics and system activity and can generate alerts when unusual conditions occur.

This architecture provides scalability, availability and security. The use of multiple application servers means that failure of one server does not necessarily stop the entire application. Auto Scaling allows the system to respond automatically to changing demand during major events.

Task 5 — Feedback

Example Submission

Person asked: Friend/Family member

Question/Feedback:

"Why do you need two application servers instead of one?"

My Response:

"I explained that two servers provide redundancy and allow a load balancer to distribute traffic between them. If one server fails, the other can continue serving users, and additional servers can be added when traffic increases."